

PySpark

Introduction to pyspark :

- PySpark is the Python API for Apache Spark, a powerful open-source framework used for large-scale data processing.
- It allows developers and data engineers to process massive datasets efficiently using distributed computing.

Why Pyspark ?

Traditional tools like Python (pandas) process data on a single machine, which leads to:

- Memory limitations
- Slow processing for large datasets

PySpark solves this by:

- Distributing data across multiple machines
- Processing data in parallel
- Providing faster and scalable performance

Where PySpark is Used ?

- Data Engineering pipelines
- Big Data analytics
- Cloud-based data platforms
- Real-time systems

Industries :

- Banking & Finance
- E-commerce
- Healthcare
- Telecom
- Technology companies

When to Use PySpark ?

Use PySpark when:

- Data is very large (GBs to TBs)
- Fast processing is required
- Distributed systems are involved
- Real-time processing is needed

Avoid PySpark when:

- Data is small
- Simple scripts are sufficient
- No distributed environment is available

Advantages :

- Handles large-scale data
- High performance (in-memory processing)
- Scalable
- Fault-tolerant
- Supports SQL, streaming, and machine learning
- Works with cloud platforms

PySpark vs Pandas

Feature	PySpark	Pandas
Data Size	Large (TBs)	Small (MBs/GBs)
Processing	Distributed	Single machine
Speed	High	Limited
Complexity	High	Low

Real-World Use Cases :

1. ETL (Extract, Transform, Load) pipelines
2. Log analysis
3. Fraud detection
4. Recommendation systems
5. Real-time dashboards

Disadvantages :

- Requires cluster setup
- More complex than basic Python tools
- Debugging can be difficult
- Higher resource usage